

PC2174A: Mathematical Methods in Physics I

Assignment 6

1. [20 pts] Given that $\mathbf{A}$ and $\mathbf{B}$ are arbitrary vectors, prove the following vector identity using Kronecker delta and Levi-Civita symbols:

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{A} \times \mathbf{B}) = |\mathbf{A}|^2 |\mathbf{B}|^2 - (\mathbf{A} \cdot \mathbf{B})^2.$$

Answer:

Expressing the cross product with the Levi-Civita symbol gives:

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{A} \times \mathbf{B}) = \sum_k \left(\sum_{i,j} \epsilon_{ijk} a_i b_j \sum_{l,m} \epsilon_{lmk} a_l b_m \right) = \sum_{i,j,k,l,m} \epsilon_{ijk} \epsilon_{lmk} a_i b_j a_l b_m \quad (1)$$

Knowing that:

$$\sum_k \epsilon_{ijk} \epsilon_{lmk} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl} \quad (2)$$

As a result, the original equation turns to

$$\sum_{i,j,l,m} (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) a_i b_j a_l b_m = \sum_l a_l^2 \sum_m b_m^2 - \sum_m a_m b_m \sum_l a_l b_l = |\mathbf{A}|^2 |\mathbf{B}|^2 - (\mathbf{A} \cdot \mathbf{B})^2 \quad (3)$$

2. [25 pts] In classical mechanics, the angular momentum about the origin of a particle of mass m moving with velocity $\mathbf{v}$ is defined as $\mathbf{L} = \mathbf{r} \times m\mathbf{v}$ where $\mathbf{r}$ is the position vector of the particle. Now consider a rigid collection of particles rotating about an axis through the origin, the angular velocity of the collection being represented by $\boldsymbol{\omega}$ where the direction of $\boldsymbol{\omega}$ corresponds to the axis of rotation.

- (a) The velocity of the i th particle is given by $\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i$, show that the total angular momentum about the origin is

$$\mathbf{L} = \sum_i m_i [r_i^2 \boldsymbol{\omega} - (\mathbf{r}_i \cdot \boldsymbol{\omega}) \mathbf{r}_i].$$

- (b) Show further that the component of $\mathbf{L}$ along the axis of rotation can be written as $I\boldsymbol{\omega}$, where I , the moment of inertia of the collection about the axis of rotation, is given by

$$I = \sum_i m_i \rho_i^2.$$

Interpret ρ_i geometrically.

- (c) Prove that the total kinetic energy of the particles is

$$K = \frac{1}{2} I \omega^2.$$

Answer:

- (a) The total angular momentum is

$$\mathbf{L} = \sum_i m_i \mathbf{r}_i \times \mathbf{v}_i = \sum_i m_i \mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i) = \sum_i m_i (\boldsymbol{\omega} (\mathbf{r}_i \cdot \mathbf{r}_i) - \mathbf{r}_i (\boldsymbol{\omega} \cdot \mathbf{r}_i)) \quad (4)$$

- (b)

$$I\boldsymbol{\omega} = \mathbf{L} \cdot \hat{\mathbf{e}}_{\boldsymbol{\omega}} = \mathbf{L} \cdot \frac{\boldsymbol{\omega}}{|\boldsymbol{\omega}|} = \sum_i m_i \left[r_i^2 \omega - \frac{(\mathbf{r}_i \cdot \boldsymbol{\omega})^2}{|\boldsymbol{\omega}|} \right] \quad (5)$$

As a result, the moment of inertia is

$$I = \sum_i m_i \left[r_i^2 - \frac{(\mathbf{r}_i \cdot \boldsymbol{\omega})^2}{|\boldsymbol{\omega}|^2} \right] = \sum_i m_i [r_i^2 - (\mathbf{r}_i \cdot \hat{\mathbf{e}}_{\boldsymbol{\omega}})^2] \quad (6)$$

Obviously,

$$\rho_i^2 = r_i^2 - (\mathbf{r}_i \cdot \hat{\mathbf{e}}_\omega)^2 = (\mathbf{r}_i \cdot \hat{\mathbf{e}}_{\perp\omega})^2 \quad (7)$$

can be considered as the projection of the displacement vector onto the direction perpendicular to the axis of rotation.

(c)

$$K = \frac{1}{2} \sum_i m_i v_i^2 = \frac{1}{2} \sum_i m_i (\omega \times \mathbf{r}_i)^2 \quad (8)$$

Applying the conclusion we deriving in question 1:

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{A} \times \mathbf{B}) = |\mathbf{A}|^2 |\mathbf{B}|^2 - (\mathbf{A} \cdot \mathbf{B})^2$$

We can get

$$K = \frac{1}{2} \sum_i m_i \omega^2 (r_i^2 - (\mathbf{r}_i \cdot \hat{\mathbf{e}}_\omega)^2) = \frac{1}{2} I \omega^2 \quad (9)$$

3. [30 pts] A particle moves with uniform speed, $v(t) = \text{constant}$, along the curve

$$\rho(t) = k [1 + \cos \phi(t)],$$

(a cardioid). Here, k is a positive constant, and ρ and ϕ are polar coordinates. Find the magnitude of the acceleration in terms of v , k and ϕ .

Answer:

the velocity and acceleration in polar coordinates is

$$\mathbf{v} = \dot{r} \hat{\mathbf{e}}_r + r \dot{\theta} \hat{\mathbf{e}}_\theta \quad (10)$$

$$\mathbf{a} = (\ddot{r} - r \dot{\theta}^2) \hat{\mathbf{e}}_r + (2\dot{r} \dot{\theta} + r \ddot{\theta}) \hat{\mathbf{e}}_\theta \quad (11)$$

The velocity of the particle moving alone the curve is

$$\mathbf{v} = -k \sin \phi \dot{\phi} \hat{\mathbf{e}}_\rho + k [1 + \cos \phi] \dot{\phi} \hat{\mathbf{e}}_\phi \quad (12)$$

Since the $|\mathbf{v}|$ is constant, $\dot{\phi}$ can be derived:

$$\begin{aligned} k^2 2 \sin^2 \phi \dot{\phi}^2 + k^2 [1 + \cos \phi]^2 \dot{\phi}^2 &= k^2 (2 + 2 \cos \phi) \dot{\phi}^2 = v^2 \\ \Rightarrow \dot{\phi} &= \frac{v}{\sqrt{2k} \sqrt{1 + \cos \phi}} \end{aligned} \quad (13)$$

To derive the magnitude of acceleration, the differentiation of ρ and ϕ should be worked out:

$$\ddot{\rho} = -k \cos \phi \dot{\phi}^2 - k \sin \phi \ddot{\phi} = \frac{v^2 \sin^2 \phi - 2 \cos \phi - 2}{4k (1 + \cos \phi)^2} \quad (14)$$

$$\ddot{\phi} = \frac{d}{d\phi} \left(\frac{d\phi}{dt} \right) \left(\frac{d\phi}{dt} \right) = \frac{v^2}{4k^2} \frac{\sin \phi}{(1 + \cos \phi)^2} \quad (15)$$

The other terms in acceleration:

$$\rho \dot{\phi}^2 = \frac{v^2}{2k}, \quad 2\dot{\phi} \ddot{\phi} = -\frac{v^2}{2k} \frac{2 \sin \phi}{1 + \cos \phi}, \quad \rho \ddot{\phi} = \frac{v^2}{4k} \frac{\sin \phi}{1 + \cos \phi} \quad (16)$$

Substitute above equations into equation (11) and we derive:

$$a_\rho = \frac{3v^2 \sin^2 \phi - 2 \cos \phi - 2}{4k (1 + \cos \phi)^2}, \quad a_\phi = -\frac{3v^2}{4k} \frac{\sin \phi}{1 + \cos \phi} \quad (17)$$

As a result, the magnitude of acceleration is:

$$|a| = \sqrt{a_\rho^2 + a_\phi^2} = \frac{3v^2}{4k} \frac{\sqrt{16 \cos^6 \frac{\phi}{2}}}{(1 + \cos \phi)^2} = \frac{3v^2}{4k |\cos \frac{\phi}{2}|} \quad (18)$$

4. [25 pts]

- (a) The force acting on a moving charged particle in a magnetic field $\mathbf{B}$ is $\mathbf{F} = q \mathbf{v} \times \mathbf{B}$ where q is the electric charge of the particle and $\mathbf{v}$ is its velocity. Suppose that a particle moves in the xy -plane with a uniform magnetic field in the z -direction. Assuming Newton's second law, $\mathbf{F} = m \frac{d\mathbf{v}}{dt}$, show that the force and velocity are perpendicular and that both have constant magnitude.
- (b) Two moving charged particles exert forces on each other because each one creates a magnetic field in which the other moves. The magnetic field $\mathbf{B}_1$ at $\mathbf{r}$ created by a charged particle q_1 at $\mathbf{r}_1$ with velocity $\mathbf{v}_1$ is given by

$$\mathbf{B}_1(\mathbf{r}) = \frac{\mu_0 q_1}{4\pi} \frac{\mathbf{v}_1 \times (\mathbf{r} - \mathbf{r}_1)}{|\mathbf{r} - \mathbf{r}_1|^3},$$

where μ_0 is the physical constant known as permeability of vacuum. Are the forces between two moving charges, q_1 and q_2 , obeying Newton's third law? If not, find the condition for which Newton's third law is still obeyed.

Answer:

(a)

$$\mathbf{v} \cdot \mathbf{F} = \mathbf{v} \cdot (q\mathbf{v} \times \mathbf{B}) = q\mathbf{v} \cdot (\mathbf{B} \times \mathbf{v}) \Rightarrow \mathbf{v} \cdot \mathbf{F} = 0 \quad (19)$$

Obviously, force and velocity are perpendicular.

$$\frac{d(v^2)}{dt} = 2\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} = \frac{2}{m} \mathbf{v} \cdot \mathbf{F} = 0 \quad (20)$$

The magnitude of velocity is a constant. Since velocity and the magnetic field are also perpendicular,

$$|F| = q|v|B_z \quad (21)$$

F is constant as well.

Q.E.D.

(b) The force exerted by particle 1 on particle 2 is

$$\begin{aligned} \mathbf{F}_{12} &= q_2 \mathbf{v}_2 \times \mathbf{B}_1(\mathbf{r}_2) = \frac{\mu_0 q_1 q_2}{4\pi |\mathbf{r}_1 - \mathbf{r}_2|^3} \mathbf{v}_2 \times [\mathbf{v}_1 \times (\mathbf{r}_2 - \mathbf{r}_1)] \\ &= \frac{\mu_0 q_1 q_2}{4\pi |\mathbf{r}_1 - \mathbf{r}_2|^3} (\mathbf{v}_1 [\mathbf{v}_2 \cdot (\mathbf{r}_2 - \mathbf{r}_1)] - (\mathbf{r}_2 - \mathbf{r}_1)(\mathbf{v}_1 \cdot \mathbf{v}_2)) \end{aligned} \quad (22)$$

The force exerted by particle 2 on particle 1 is

$$\begin{aligned} \mathbf{F}_{21} &= q_1 \mathbf{v}_1 \times \mathbf{B}_2(\mathbf{r}_1) = \frac{\mu_0 q_1 q_2}{4\pi |\mathbf{r}_1 - \mathbf{r}_2|^3} \mathbf{v}_1 \times [\mathbf{v}_2 \times (\mathbf{r}_1 - \mathbf{r}_2)] \\ &= \frac{\mu_0 q_1 q_2}{4\pi |\mathbf{r}_1 - \mathbf{r}_2|^3} (\mathbf{v}_2 [\mathbf{v}_1 \cdot (\mathbf{r}_1 - \mathbf{r}_2)] - (\mathbf{r}_1 - \mathbf{r}_2)(\mathbf{v}_2 \cdot \mathbf{v}_1)) \end{aligned} \quad (23)$$

Obviously, they are not opposite since $\mathbf{F}_{12}$ lies in the plane spanned by $\mathbf{v}_1$ and $\mathbf{r}_2 - \mathbf{r}_1$, while $\mathbf{F}_{21}$ lies in the plane spanned by $\mathbf{v}_2$ and $\mathbf{r}_2 - \mathbf{r}_1$.

Only in case when $\mathbf{v}_1 = \mathbf{v}_2$ can Newton's third law be still obeyed.

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